

APPLETON, MN, USA

WMO: 726466

Lat: 45.228N

Lon: 96.007W

Elev: 1020

StdP: 14.16

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54917

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.3	-9.5	-20.4	1.9	-13.8	-16.9	2.3	-10.1	32.7	14.2	28.9	12.1	6.0	250	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	90.4	73.4	87.8	72.4	84.1	70.2	77.7	86.0	75.5	83.1	73.5	81.0	11.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.0	136.5	82.5	72.8	126.3	80.5	71.7	121.6	79.4	42.0	85.5	39.6	82.9	37.8	81.3	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.5	24.4	20.8	-22.0	95.3	5.5	3.2	-26.0	97.5	-29.2	99.4	-32.3	101.2	-36.4	103.5		
			-22.6	81.1	5.5	2.6	-26.5	83.0	-29.7	84.6	-32.8	86.1	-36.7	88.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	13.2	16.3	29.5	44.4	57.2	67.4	71.9	68.6	61.4	46.9	32.4	18.4	(6)
	DBStd	23.10	13.37	13.27	13.19	10.30	9.10	6.58	6.03	6.05	8.58	9.73	11.10	12.47	(7)
	HDD50	4690	1141	944	645	222	36	0	0	0	10	177	535	980	(8)
	HDD65	8217	1606	1364	1102	619	273	49	14	32	168	567	978	1445	(9)
	CDD50	2537	0	0	9	55	259	522	677	577	351	80	7	0	(10)
	CDD65	589	0	0	0	2	32	121	227	144	59	4	0	0	(11)
	CDH74	5903	0	0	3	51	436	1200	2195	1297	642	78	1	0	(12)
	CDH80	1903	0	0	0	13	145	366	765	389	205	20	0	0	(13)
	WSAvg	9.0	9.4	9.8	10.0	10.7	10.6	8.7	6.9	6.5	7.7	9.3	9.5	9.3	(14)
	PrecAvg	25.4	0.7	0.7	1.3	2.5	2.8	3.9	3.8	2.9	2.8	2.4	1.1	0.6	(15)
(16) Precipitation	PrecMax	32.3	1.8	1.6	3.5	6.7	4.6	7.5	7.4	5.2	6.2	7.3	3.7	1.9	(16)
	PrecMin	16.4	0.0	0.2	0.3	0.4	0.7	1.4	1.3	0.6	0.5	0.2	0.0	0.0	(17)
	PrecStd	4.0	0.5	0.4	0.8	1.4	1.1	1.8	1.6	1.3	1.5	1.9	1.0	0.5	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	45.0	53.6	71.6	81.0	90.5	91.4	93.3	92.8	90.6	82.2	67.6	50.1		(19)
	0.4% MCWB	38.0	49.3	55.5	60.1	66.9	71.4	77.7	75.8	70.8	62.1	54.8	42.8		(20)
	2% DB	37.3	43.4	61.0	72.8	83.8	88.1	90.3	88.1	85.9	73.4	60.5	43.0		(21)
	2% MCWB	34.1	39.1	52.1	55.1	63.3	70.3	76.1	74.7	67.9	57.9	50.2	37.0		(22)
	5% DB	35.8	37.4	54.2	67.6	79.0	83.9	87.8	83.9	81.5	68.3	54.1	37.2		(23)
	5% MCWB	33.7	35.1	48.0	52.9	61.5	68.5	74.8	72.4	67.0	56.6	46.6	34.3		(24)
	10% DB	31.9	35.7	46.3	62.8	73.1	81.2	84.0	81.3	77.0	63.2	47.9	35.6		(25)
	10% MCWB	30.9	34.1	42.2	50.4	59.1	67.0	72.2	70.2	64.4	53.5	42.1	33.5		(26)
	0.4% WB	38.5	46.9	58.7	62.1	71.4	76.8	82.2	80.4	75.2	67.3	57.7	44.2		(27)
	0.4% MCDB	43.9	51.6	63.7	76.4	83.3	85.9	89.0	86.1	84.5	76.4	64.6	49.5		(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	35.1	39.5	53.1	57.8	67.2	74.4	78.6	76.9	72.6	62.5	51.1	37.6		(29)
	2% MCDB	37.4	43.1	59.8	68.5	78.1	83.1	86.6	85.1	81.9	69.2	57.6	41.4		(30)
	5% WB	33.0	36.1	47.3	54.5	64.1	71.9	76.6	74.4	70.0	57.6	46.7	35.0		(31)
	5% MCDB	35.1	38.2	53.4	64.7	73.3	80.6	84.4	81.4	77.8	65.7	53.4	37.6		(32)
	10% WB	30.4	33.3	42.4	51.5	61.4	69.3	74.4	72.1	67.1	54.3	42.6	32.6		(33)
	10% MCDB	31.5	35.2	47.4	60.1	71.6	77.5	81.6	79.2	74.4	61.8	48.0	34.8		(34)
(35) Mean Daily Temperature Range	MDBR	17.6	18.1	17.6	21.1	22.0	20.7	21.0	21.3	23.3	20.8	18.1	16.2		(35)
	5% DB	20.5	19.5	27.1	31.6	29.4	25.6	23.3	24.4	28.4	30.4	27.3	20.7		(36)
	5% WB	17.6	15.7	17.8	16.8	13.8	12.0	11.7	12.6	13.9	16.6	18.0	16.4		(37)
	MCDBR	19.2	18.0	26.0	26.9	23.6	20.8	20.5	21.8	23.6	25.4	25.5	19.1		(38)
	MCWBR	16.8	14.5	17.8	16.1	13.4	12.1	12.2	13.2	14.1	15.7	17.7	15.4		(39)
(40) Clear-Sky Solar Irradiance	taub	0.266	0.277	0.308	0.360	0.382	0.391	0.394	0.401	0.363	0.316	0.281	0.262		(40)
	taud	2.436	2.418	2.440	2.338	2.342	2.374	2.350	2.315	2.406	2.520	2.530	2.462		(41)
	Ebn at Noon	273	291	295	285	281	278	276	269	272	273	266	263		(42)
	Edh at Noon	22	27	31	37	39	37	38	38	32	25	20	19		(43)
(44) All-Sky Solar Radiation	RadAvg	561	903	1206	1513	1738	1940	2071	1724	1330	852	554	407		(44)
	RadStd	65	101	113	128	128	123	76	144	96	115	54	44		(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (52 neighbors)	N/A	N/A	N/A	N/A	-1.71	-1.64	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page